Cardiovascular System (CARD) · NPLEX-P1-CARD-12

CV-12 Exercise and the Cardiovascular System

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-12-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the two inputs that start the exercise response

A competent response contains

· Heart rate rises within the first second of exercise, too fast for circulating metabolites to explain, so the trigger is neural.

· Two inputs converge on the medullary cardiovascular centers: central command (feedforward) and the exercise pressor reflex (feedback from muscle).

· Central command comes from the motor cortex and subcortical locomotor regions, which signal the medullary centers in parallel with recruiting motor units.

· Its output scales with the effort commanded rather than the force produced, and it resets the arterial baroreflex to a higher set point so the reflex still buffers beat to beat changes without opposing the rise in pressure.

· Mechanoreflex: group III thinly myelinated afferents, excited by stretch and tension of contracting muscle, strongest in the first seconds, producing an early heart rate rise largely by reducing vagal outflow.

· Metaboreflex: group IV unmyelinated C fiber afferents, excited by H+, lactate, K+, inorganic phosphate, ATP, bradykinin, and prostaglandins, building over tens of seconds, strongest when flow is too low for metabolism, producing sympathetic activation and a rise in arterial pressure.

· The metaboreflex works as an error signal: inadequate flow lets metabolites accumulate and the reflex raises arterial pressure to push blood into the underperfused muscle.

· Observation 1: when muscles are partly weakened by a neuromuscular blocking drug, the same force takes more effort and heart rate and blood pressure rise more, which shows the response follows commanded effort. Central command also explains the anticipatory rise before a race.

· Observation 2: trapping metabolites in the arm with a cuff after a sustained handgrip keeps blood pressure high while heart rate returns toward baseline, which shows the heart rate response depends mostly on central command and the mechanoreflex.

· Both arms converge on the same brainstem centers and interact.

Distinctions that matter

· Central command is feedforward and scales with commanded effort; the exercise pressor reflex is feedback and scales with what is happening in the muscle.

· The metaboreflex raises pressure more than rate, while central command and the mechanoreflex account for most of the rate response.

Common misconceptions

· Saying accumulating metabolites start the response. They build over tens of seconds, while heart rate rises within the first second.

· Saying central command switches the baroreflex off. It resets the set point upward, so buffering continues around a higher pressure.

Where it is taught: Section C01, opening paragraphs, the exercise pressor reflex table, and the metaboreflex paragraph of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C01-BD2

Brain dump 2 · integration and application: Why the rate rise is vagal first and sympathetic second

A competent response contains

· Vagal acetylcholine acts on M2 receptors; Gi beta-gamma subunits open IKACh channels directly, and acetylcholinesterase clears acetylcholine within milliseconds, so stopping vagal firing raises heart rate within one or two beats.

· Norepinephrine acts on beta-1 receptors through Gs, cAMP, and protein kinase A, and is cleared mainly by reuptake, so its effects build and fade over seconds.

· Vagal withdrawal produces most of the early rise, up to roughly 100 bpm in a young adult.

· Above that, sympathetic drive raises rate further, increases contractility, and constricts veins and arterioles in inactive tissues.

· Venous return rises through sympathetic venoconstriction of the splanchnic and cutaneous veins, which raises mean systemic filling pressure, the skeletal muscle pump driving blood past venous valves toward the heart, and the respiratory pump as deeper breathing lowers intrathoracic pressure.

· Greater filling raises stroke volume through the Frank-Starling mechanism.

· Transplanted heart: denervated, so the resting rate sits near the intrinsic SA node rate and barely changes at onset; cardiac output rises first through the Frank-Starling response, and rate climbs over minutes as circulating catecholamines rise, to a lower peak than normal.

· Beta-1 blockade blunts the sympathetic part of the rate response but spares vagal withdrawal, so the early rise is largely preserved and the later rise is blunted.

· Heart failure: an exaggerated exercise pressor reflex adds excess sympathetic vasoconstriction during exercise.

Distinctions that matter

· Speed of each limb follows its signaling and its clearance: a directly gated channel with enzymatic clearance acts in milliseconds, a second messenger cascade with reuptake acts over seconds.

· Raising cardiac output needs venous return to rise with it, so the venous and pump mechanisms are part of the response rather than a side note.

Common misconceptions

· Treating the early rise in heart rate as sympathetic. It is mostly loss of vagal restraint.

· Expecting a denervated heart to be unable to raise cardiac output. It raises stroke volume by the Frank-Starling mechanism and raises rate later through circulating catecholamines.

Where it is taught: Section C01, paragraphs on vagal withdrawal, venous return, and the final paragraph on denervation, blockade, and heart failure of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C01-RT01

Retrieval exercise: Rebuild the onset sequence

A competent response contains

· Step 1: central command signals the medullary centers as motor units are recruited.

· Step 2: vagal outflow falls and heart rate rises within one or two beats, because IKACh channels close and acetylcholine is cleared within milliseconds.

· Step 3: group III mechanoreceptors reinforce the early heart rate rise.

· Step 4: sympathetic outflow builds over seconds and raises rate, raises contractility, and raises venous tone at the same time.

· Step 5: the muscle pump drives blood past venous valves toward the heart and the respiratory pump lowers intrathoracic pressure, so venous return and preload rise.

· Step 6: stroke volume and cardiac output rise as active arterioles dilate.

· Step 7: if flow lags metabolism, group IV afferents trigger the metaboreflex and raise arterial pressure.

· Central command starts the response and muscle feedback matches it to the working muscle.

Distinctions that matter

· The rate response and the stroke volume response arrive on different timescales: rate within a beat or two, stroke volume once venous return has risen.

· The metaboreflex is a correction that only comes in when perfusion is not keeping up, not a routine step in every bout.

Common misconceptions

· Putting sympathetic activation first. Vagal withdrawal precedes it.

· Leaving out the pumps and assuming a stronger heart alone raises cardiac output. Output cannot rise unless venous return rises with it.

Where it is taught: Section C01, flow diagram on the cardiovascular response at the onset of exercise of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C02-BD1

Brain dump 1 · foundational retrieval: Write the Fick principle and fill in the numbers

A competent response contains

· Fick principle: VO2 equals cardiac output times the arteriovenous O2 difference, which is arterial minus mixed venous O2 content. It holds because O2 is not stored.

· Arterial O2 content is about 1.34 mL O2 per gram of hemoglobin times hemoglobin concentration times saturation, plus 0.003 mL/dL per mmHg of dissolved PO2, giving about 20 mL/dL at a hemoglobin of 15 g/dL and a saturation of 98%.

· Healthy lungs hold arterial content near 20 mL/dL throughout exercise, so the arteriovenous difference widens from the venous side.

· Heart rate: about 70/min at rest and about 50/min in an athlete, about 195/min at maximum untrained, about 185 to 195/min in an elite athlete.

· Stroke volume: about 70 mL at rest and about 100 mL in an athlete, about 110 mL at maximum untrained, about 160 mL or more in an elite athlete.

· Cardiac output: about 5 L/min at rest, about 20 to 22 L/min at maximum untrained, about 30 to 35 L/min in an elite athlete.

· Mixed venous O2 content: about 15 mL/dL at rest (saturation about 75%), about 5 mL/dL at maximum untrained, about 3 to 4 mL/dL in an elite athlete.

· Arteriovenous O2 difference: about 5 mL/dL at rest, about 15 mL/dL at maximum untrained, about 16 to 17 mL/dL in an elite athlete.

· VO2: about 250 mL/min at rest (3.5 mL/kg/min, 1 MET), about 3 L/min (35 to 45 mL/kg/min) at maximum untrained, about 5 to 6 L/min (70 to 85 mL/kg/min) in an elite athlete.

· Extraction rises because metabolites recruit closed capillaries, enlarging the exchange surface and shortening diffusion distance; tissue PO2 falls onto the steep part of the dissociation curve; and higher PCO2, H+, and temperature shift the curve to the right.

· The right shift from CO2 and H+ is the Bohr effect.

Distinctions that matter

· Arterial content stays about the same while venous content falls, so the widening difference is a venous event.

· Capillary recruitment changes the exchange surface; the right shift changes hemoglobin's affinity. They are separate contributions to the same rise in extraction.

Common misconceptions

· Expecting arterial O2 content or saturation to rise with exercise in a healthy person. It is already nearly fully saturated at rest.

· Treating dissolved O2 as negligible to the point of leaving it out of the content equation; it is small but part of the calculation.

Where it is taught: Section C02, opening paragraphs, the table of typical values at rest and maximum, and the extraction paragraph of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C02-BD2

Brain dump 2 · integration and application: Work VO2max from the numbers, then take the hemoglobin away

A competent response contains

· Rest: 70/min times 70 mL is about 4.9 L/min; the difference is 20 minus 15, or 5 mL/dL, which is 50 mL/L, so VO2 is about 245 mL/min, roughly 3.3 mL/kg/min.

· Maximum: 195/min times 110 mL is about 21.5 L/min; the difference is 20 minus 5, or 15 mL/dL, which is 150 mL/L, so VO2max is about 3,200 mL/min, about 43 mL/kg/min.

· Flow rose about 4.4 fold and extraction about threefold.

· With a hemoglobin of 9 g/dL, arterial content is about 12 mL/dL, so even at a venous content of 4 mL/dL the maximal difference is 8 mL/dL, which is 80 mL/L.

· Holding a VO2 of about 2,400 mL/min would then need about 30 L/min of cardiac output, which is beyond the untrained maximum, so VO2max falls.

· Because each liter of blood carries less O2, heart rate is higher at any fixed submaximal workload.

· In a healthy person at sea level, VO2max is limited mainly by maximal cardiac output rather than by the lungs or the muscle: raising arterial O2 content raises VO2max and lowering it reduces VO2max, so maximal stroke volume is the main determinant.

· Stroke volume plateaus at about 40% to 60% of VO2max in untrained people because shortened diastole limits filling, so further rises in cardiac output come from heart rate; trained endurance athletes may keep raising stroke volume.

· Cardiac output rises about 5 to 6 L/min for each 1 L/min rise in VO2.

Distinctions that matter

· Anemia lowers the ceiling on the arteriovenous difference rather than the ability to extract: venous content still falls, but arterial content started lower.

· VO2 at a fixed submaximal workload is set by the work, so anemia shows up as a higher heart rate at that workload rather than a lower VO2.

Common misconceptions

· Mixing units: content in mL/dL must be converted to mL/L before multiplying by cardiac output in L/min.

· Naming the lungs as the limit to VO2max in healthy people at sea level. Arterial saturation is maintained, and maximal cardiac output is the limiting step.

Where it is taught: Section C02, worked example on the Fick equation and the paragraphs on heart rate, stroke volume, and VO2max of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C03-BD1

Brain dump 1 · foundational retrieval: Rebuild the distribution of cardiac output

A competent response contains

· Skeletal muscle: about 1.2 L/min (20%) at rest, about 22 L/min (about 87%) at maximum, controlled by metabolites and nitric oxide.

· Coronary: about 0.25 L/min (4%) at rest, about 1 L/min (4%) at maximum, controlled metabolically and mainly by adenosine.

· Brain: about 0.75 L/min (13%) at rest, about 0.75 L/min (3%) at maximum, controlled by autoregulation and PCO2.

· Skin: about 0.5 L/min (9%) at rest, about 0.6 L/min at maximum and higher during heavy submaximal work, controlled by hypothalamic thermoregulation.

· Splanchnic: about 1.4 L/min (24%) at rest, about 0.3 L/min at maximum, controlled by sympathetic constriction.

· Kidneys: about 1.1 L/min (19%) at rest, about 0.25 L/min at maximum, controlled by sympathetic constriction and angiotensin II.

· Muscle flow rises from about 3 to 4 mL/100 g/min at rest to about 50 to 80 mL/100 g/min in strenuous exercise, up to about 15 to 25 times resting.

· Interstitial K+ from repeated action potentials hyperpolarizes vascular smooth muscle through inward rectifier K+ channels and the Na+/K+-ATPase.

· Adenosine, inorganic phosphate, H+, CO2, and low PO2 add to the dilation; low PO2 makes red cells release ATP, which stimulates endothelial nitric oxide.

· Shear stress releases nitric oxide upstream, spreading dilation to the feed arteries, and blocking any single mediator only partly reduces the response.

· Extra cardiac output goes to active muscle, the heart, and, as body temperature rises, the skin.

Distinctions that matter

· Coronary flow quadruples while its percentage share stays about 4%, so a constant share is not a constant flow.

· Brain flow is held constant in absolute terms, so its percentage share falls as total output rises.

Common misconceptions

· Reading a falling percentage share as a falling flow. Cerebral flow is unchanged in L/min.

· Naming one mediator as the cause of muscle hyperemia. Several act together, which is why blocking one has only a partial effect.

Where it is taught: Section C03, distribution table and the paragraph on active skeletal muscle of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C03-BD2

Brain dump 2 · integration and application: Why total resistance falls while some beds constrict

A competent response contains

· Sympathetic nerves keep firing during exercise, but local metabolites blunt alpha-adrenergic constriction in active muscle, especially through alpha-2 receptors, which is functional sympatholysis, so inactive beds constrict while active muscle is protected.

· Some residual constriction is needed, because the dilating capacity of all skeletal muscle exceeds maximal cardiac output.

· Because the beds lie in parallel, their conductances add, so total peripheral resistance falls steeply in dynamic exercise even though other beds constrict.

· The myocardium already extracts about 70% to 75% of delivered O2 at rest, so higher demand has to be met by flow, which rises about three- to fourfold through mainly adenosine-mediated dilation.

· A coronary stenosis that is silent at rest therefore causes angina on exertion, because exercise raises myocardial O2 demand and the stenosis limits the required rise in flow.

· Cerebral flow stays nearly constant through autoregulation and PCO2 sensitivity.

· Splanchnic and renal flow fall to about one quarter of resting values at maximal effort, and splanchnic venoconstriction moves reservoir blood centrally.

· Renal filtration fraction rises because renal blood flow falls more than GFR does.

Distinctions that matter

· Sympathetic outflow rising everywhere and flow rising only in active muscle are reconciled by local control at the arteriole, not by selective nerve traffic.

· A bed meeting demand by extraction has no reserve left, which is why the coronary bed must answer with flow.

Common misconceptions

· Saying sympathetic outflow to active muscle is switched off. It continues, and its vasoconstrictor effect is blunted locally.

· Expecting total peripheral resistance to rise because most beds constrict. Parallel conductances add, and the enormous muscle conductance dominates.

Where it is taught: Section C03, functional sympatholysis paragraph, the flow diagram, and the paragraph on the other organs of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C03-RT01

Retrieval exercise: Rebuild how flow concentrates in working muscle

A competent response contains

· Step 1: sympathetic outflow rises to all vascular beds.

· Step 2: gut, kidney, and inactive muscle arterioles constrict through alpha receptors.

· Step 3: in active muscle, K+, adenosine, H+, CO2, low PO2, and nitric oxide dilate the arterioles.

· Step 4: the same metabolites blunt alpha constriction, which is functional sympatholysis, acting especially at alpha-2 receptors.

· Step 5: resistance falls in active muscle and rises in inactive beds.

· Step 6: the parallel beds share one arterial pressure, so flow shifts to active muscle.

· Flow through a bed equals the shared perfusion pressure divided by that bed's resistance, so each bed's share depends on its resistance relative to the others.

Distinctions that matter

· Constriction elsewhere and dilation in muscle both push flow the same way, but only the dilation explains the fall in total peripheral resistance.

· A shared perfusion pressure is what makes resistance the deciding variable for each bed's share.

Common misconceptions

· Treating the redistribution as a central decision about where to send blood. The brainstem raises outflow globally, and the arterioles settle the distribution locally.

· Omitting the shared arterial pressure from the explanation, which leaves no reason why one bed's dilation changes another bed's flow.

Where it is taught: Section C03, flow diagram on how flow concentrates in working muscle of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C03-PR01

Prediction exercise: Predict what heat does to a steady run

A competent response contains

· Skin blood flow rises on the hot day, because the hypothalamus withdraws cutaneous vasoconstrictor tone and activates sympathetic cholinergic fibers that cause active vasodilation and sweating, carrying heat to the skin.

· Blood available to fill the heart falls, because blood pools in compliant skin veins and sweat loss reduces plasma volume.

· Stroke volume falls over the 60 minutes as filling falls.

· Heart rate drifts upward to hold cardiac output, which is cardiovascular drift.

· Flow to the leg muscles falls, because muscle now shares cardiac output with skin, which lowers the sustainable workload.

· At maximal effort skin flow falls again, which favors muscle but limits heat loss, so core temperature rises further; above 40 °C (104 °F) core temperature reaches the range of exertional heat stroke.

· Core temperature rises toward 38 to 39 °C (100.4 to 102.2 °F) in sustained heavy exercise even without heat stress.

Distinctions that matter

· Cardiovascular drift is a fall in stroke volume with a compensating rise in heart rate, so cardiac output is roughly held rather than raised.

· Skin flow rises during heavy submaximal work and falls again at maximal effort, so the direction depends on the intensity.

Common misconceptions

· Reading a rising heart rate at a constant workload as rising cardiac output. It is compensation for a falling stroke volume.

· Assuming skin flow keeps rising as intensity rises. At maximal effort it is sacrificed to muscle perfusion.

Where it is taught: Section C03, skin blood flow paragraph and the stop and predict box of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C04-BD1

Brain dump 1 · foundational retrieval: Pressure in dynamic exercise against pressure in static exercise

A competent response contains

· Mean arterial pressure equals cardiac output times total peripheral resistance, ignoring the small central venous pressure.

· Dynamic exercise: heart rate rises largely with work rate, stroke volume rises, cardiac output rises severalfold, total peripheral resistance falls, systolic pressure rises progressively to about 160 to 200 mmHg at peak, diastolic pressure stays about the same or falls slightly, and pulse pressure widens.

· Systolic pressure rises in dynamic exercise because a larger stroke volume is ejected faster; diastolic pressure holds or falls because blood runs off through dilated muscle beds.

· The main pressor drive in dynamic exercise is central command with the mechanoreflex.

· Static exercise: heart rate rises moderately, stroke volume changes little or falls slightly, cardiac output rises modestly and mainly through heart rate, total peripheral resistance is unchanged or rises, and systolic and diastolic pressure both rise sharply.

· The main pressor drive in static exercise is the metaboreflex with central command.

· A sustained isometric contraction raises intramuscular pressure and compresses the muscle's own vessels, reducing flow above a moderate fraction of maximal force and nearly stopping it at high force, so metabolites accumulate and drive a strong sympathetic pressor response graded by relative force and duration.

· A Valsalva maneuver during maximal lifting adds intrathoracic pressure to the aorta.

· Dynamic exercise imposes a volume load and the pressure-volume loop widens more than it rises; static exercise imposes a pressure load and the loop becomes taller.

· By the Laplace relationship, wall stress is proportional to pressure times radius divided by wall thickness, so a pressure load raises wall stress and O2 demand.

· Arm exercise at the same VO2 raises heart rate and pressure more than leg exercise, because less muscle dilates while the inactive leg beds stay constricted.

· Most real activities mix the two patterns; rowing adds a large static component to a volume load.

Distinctions that matter

· Both forms raise cardiac output, and the direction of the resistance change is what separates their pressure responses.

· Pulse pressure widens in dynamic exercise because diastolic pressure holds while systolic rises; in static exercise both rise, so pulse pressure changes much less.

Common misconceptions

· Expecting diastolic pressure to rise in dynamic exercise. Runoff through dilated muscle beds holds it about the same or lowers it slightly.

· Treating a handgrip as a mild version of running. It loads the ventricle in the opposite way, with pressure rather than volume.

Where it is taught: Section C04, dynamic and static exercise paragraphs and the table of acute responses of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C04-BD2

Brain dump 2 · integration and application: Read pressures and work out resistance

A competent response contains

· Mean arterial pressure equals diastolic pressure plus one third of the pulse pressure; total peripheral resistance equals mean arterial pressure divided by cardiac output.

· Rest: 120/80 mmHg gives a pulse pressure of 40 and a mean arterial pressure of about 93 mmHg; 93 divided by 5 gives a total peripheral resistance of about 19 mmHg per L/min.

· Running: 180/76 mmHg gives a pulse pressure of 104 and a mean arterial pressure of about 111 mmHg; 111 divided by 20 gives about 5.5. Cardiac output quadrupled while mean arterial pressure rose only about 17 mmHg, because resistance fell to under one third of resting.

· Handgrip: 165/110 mmHg gives a pulse pressure of 55 and a mean arterial pressure of about 128 mmHg; 128 divided by 6.5 gives about 20. Cardiac output rose about 30%, yet mean arterial pressure rose about 35 mmHg because resistance did not fall.

· The one third formula is least accurate at high heart rates, because diastole shortens, so true running mean arterial pressure lies nearer the midpoint of systolic and diastolic, which still leaves total peripheral resistance far below resting.

· The rate-pressure product, heart rate multiplied by systolic pressure, indexes myocardial O2 demand.

· Pressure work costs more O2 than volume work, so a static effort can raise the rate-pressure product and provoke ischemia at a lower heart rate than running.

· A fall in systolic pressure as dynamic workload rises is abnormal and means cardiac output cannot offset the falling resistance, as in severe left ventricular dysfunction, extensive ischemia, or severe aortic stenosis.

· An exaggerated rise suggests resistance is not falling and may precede resting hypertension.

Distinctions that matter

· Mean arterial pressure and total peripheral resistance move independently: pressure can rise modestly while resistance falls steeply, because cardiac output is the third term.

· The rate-pressure product reflects demand, while coronary flow reflects supply, which is why pressure work is the more provocative load.

Common misconceptions

· Reading a higher pressure as a higher resistance. During running, pressure is up and resistance is sharply down.

· Applying the one third formula without comment at a heart rate of 180/min, where the relative duration of diastole has changed.

Where it is taught: Section C04, worked example on calculating total peripheral resistance and the final paragraph on the rate-pressure product and abnormal responses of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C04-CP01

Comparison table: Dynamic against static exercise

A competent response contains

· Heart rate: large rise with work rate in dynamic exercise, moderate rise in static exercise.

· Stroke volume: rises in dynamic exercise, little change or a slight fall in static exercise.

· Cardiac output: large rise in dynamic exercise, modest rise in static exercise and mainly from heart rate.

· Total peripheral resistance: falls in dynamic exercise, unchanged or higher in static exercise.

· Systolic pressure: rises progressively in dynamic exercise, rises sharply in static exercise.

· Diastolic pressure: unchanged or slightly lower in dynamic exercise, so pulse pressure widens; rises sharply in static exercise, so pulse pressure widens much less.

· Main pressor drive: central command and the mechanoreflex in dynamic exercise, the metaboreflex and central command in static exercise.

· Load on the left ventricle: volume load in dynamic exercise, with the pressure-volume loop widening more than it rises; pressure load in static exercise, with the loop becoming taller.

· A sustained handgrip raises afterload, so the murmurs of mitral regurgitation, aortic regurgitation, and ventricular septal defect get louder, while the murmur of hypertrophic cardiomyopathy softens.

Distinctions that matter

· Volume load and pressure load are the summary of the whole table, and the direction of the resistance change is what produces them.

· Raising afterload increases regurgitant and shunt murmurs because more blood takes the low pressure route, and softens the obstructive murmur of hypertrophic cardiomyopathy because the ventricle holds a larger cavity.

Common misconceptions

· Writing that cardiac output barely changes in static exercise. It rises modestly, mainly through heart rate.

· Expecting every murmur to behave the same way with handgrip. The obstructive murmur of hypertrophic cardiomyopathy moves in the opposite direction.

Where it is taught: Section C04, table of acute responses and the final paragraph on handgrip and murmurs of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C05-BD1

Brain dump 1 · foundational retrieval: Eccentric and concentric remodeling, and the trained heart

A competent response contains

· Adult myocytes rarely divide, so the heart adapts to a chronic load by enlarging existing cells.

· Eccentric hypertrophy adds sarcomeres in series, enlarging the cavity with proportional wall thickening, and is the response to a volume load.

· Concentric hypertrophy adds sarcomeres in parallel, thickening the wall relative to cavity size, and is the response to a pressure load.

· Both patterns bring wall stress back toward normal, which follows from the Laplace relationship.

· The endurance-trained left ventricle is remodeled eccentrically, with mild symmetric wall thickening, a normal wall-to-cavity ratio, an enlarged right ventricle and atria, and normal or enhanced diastolic filling.

· Plasma volume expands within days through renin-angiotensin-aldosterone, vasopressin, and albumin, faster than red cell mass, so hemoglobin concentration may fall slightly by dilution rather than from anemia.

· With resting cardiac output still about 5 L/min, a stroke volume of about 100 mL needs only about 50 bpm.

· The slowing comes mainly from vagal tone, with a contribution from a lower intrinsic SA node rate.

· Vagal tone also explains benign sinus arrhythmia, first degree AV block, and Mobitz I block, all of which disappear with exercise.

· At the same submaximal workload, the trained person reaches the same VO2 and cardiac output with a lower heart rate, a larger stroke volume, and less lactate.

· At maximum, cardiac output, extraction, and VO2max rise, but maximal heart rate does not; greater muscle capillary and mitochondrial density raise extraction and fat oxidation, and VO2max typically rises about 15% to 20% in previously sedentary people.

Distinctions that matter

· Training raises stroke volume and VO2max while leaving resting cardiac output and maximal heart rate about the same.

· A slightly lower hemoglobin concentration in a trained athlete is dilution from plasma volume expansion, not iron deficiency.

Common misconceptions

· Explaining athletic bradycardia as a weaker heart. The rate is lower because each beat ejects more.

· Reading first degree AV block or Mobitz I in a trained athlete as disease. Both are vagally mediated and resolve with exercise.

Where it is taught: Section C05, opening paragraph, the endurance-trained heart paragraphs, and the flow diagram from volume load to higher VO2max of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C05-BD2

Brain dump 2 · integration and application: Separate physiologic growth from sarcomere disease

A competent response contains

· Physiologic hypertrophy is driven by insulin-like growth factor 1 through PI3K and Akt, with matching capillary growth and no fibrosis.

· Pathologic hypertrophy is driven by wall stress and neurohormones, namely angiotensin II, endothelin 1, and norepinephrine at alpha-1 receptors, through Gq-coupled receptors and calcineurin and NFAT signaling.

· Pathologic growth reactivates fetal genes, including beta-myosin heavy chain and the natriuretic peptides, and brings fibrosis, capillary insufficiency, and diastolic and then systolic dysfunction.

· Hypertrophic cardiomyopathy is usually autosomal dominant with variable penetrance, caused by sarcomere gene mutations, most often MYH7 for beta-myosin heavy chain and MYBPC3 for cardiac myosin binding protein C.

· Exertional dyspnea: the stiff ventricle fills poorly, so filling pressures are high.

· Angina with normal epicardial arteries: a large muscle mass is supplied through diseased intramural arterioles.

· Sudden death in a young athlete: myocyte disarray with fibrosis is a substrate for reentrant ventricular arrhythmias.

· Dynamic outflow obstruction comes from the septum together with systolic anterior motion of the mitral valve, and it worsens with lower preload (standing, Valsalva strain, dehydration), lower afterload, greater contractility, or tachycardia, because each makes the ventricle smaller or more forceful.

· The murmur therefore gets louder with standing or Valsalva and softer with squatting or handgrip, which raise preload and afterload and enlarge the cavity.

· After about three months of detraining the athlete's heart regresses, while hypertrophic cardiomyopathy does not, which is why detraining separates them.

· The same muscle mass is physiologic or pathologic according to the stimulus and the signaling pathway behind it.

Distinctions that matter

· Both conditions enlarge myocytes, and the pathway (IGF-1 with PI3K and Akt against Gq with calcineurin and NFAT) decides whether capillaries and interstitium keep pace.

· Anything that shrinks the cavity worsens the gradient, so maneuvers are judged by their effect on ventricular size rather than on pressure alone.

Common misconceptions

· Expecting angina to mean epicardial coronary disease. In hypertrophic cardiomyopathy the large mass and diseased intramural arterioles are enough.

· Predicting that handgrip makes every murmur louder. It raises afterload, enlarges the cavity, and softens this murmur.

Where it is taught: Section C05, paragraphs on physiologic versus pathologic hypertrophy, on hypertrophic cardiomyopathy, on dynamic obstruction, and the stop and predict box of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C05-CP01

Comparison table: Athlete's heart against hypertrophic cardiomyopathy

A competent response contains

· Wall thickness: symmetric and usually 12 mm or less in the athlete; often asymmetric septal and 15 mm or more in hypertrophic cardiomyopathy.

· Cavity: enlarged in the athlete, often above 55 mm; small or normal in hypertrophic cardiomyopathy, often below 45 mm.

· Diastolic function: normal or enhanced in the athlete; impaired relaxation in hypertrophic cardiomyopathy.

· Histology: orderly myocytes with no fibrosis in the athlete; myocyte disarray with fibrosis in hypertrophic cardiomyopathy.

· Outflow tract: no obstruction in the athlete; dynamic obstruction in many patients with hypertrophic cardiomyopathy.

· Electrocardiogram: bradycardia with isolated high voltage in the athlete; deep T wave inversion and pathologic Q waves in hypertrophic cardiomyopathy.

· Peak VO2: high in the athlete; normal or reduced in hypertrophic cardiomyopathy.

· Detraining for about three months: the athlete's heart regresses, hypertrophic cardiomyopathy does not.

· Family history: usually negative in the athlete; often hypertrophic cardiomyopathy or sudden death in affected families.

· Thresholds are approximate, and in the 13 to 15 mm gray zone the features are weighed together.

Distinctions that matter

· Wall thickness alone does not separate the two in the gray zone; the cavity size, diastolic filling, and family history carry the decision.

· A large cavity with proportional thickening is eccentric remodeling, while a small cavity with a thick asymmetric wall is not a training pattern at all.

Common misconceptions

· Using high QRS voltage as evidence of disease. Isolated high voltage is common in trained athletes.

· Treating the thresholds as fixed cutoffs rather than approximations that are weighed together.

Where it is taught: Section C05, athlete's heart against hypertrophic cardiomyopathy table of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C05-CA01

Clinical application: Near syncope in a young athlete during a hot race

A competent response contains

· Recognize: hypertrophic cardiomyopathy from sarcomere protein gene mutations, most often MYH7 for beta-myosin heavy chain and MYBPC3 for cardiac myosin binding protein C, usually autosomal dominant with variable penetrance.

· Explain: the stiff ventricle fills poorly, so filling pressures rise and exertional dyspnea follows; a large muscle mass supplied through diseased intramural arterioles causes angina even when the epicardial arteries are normal, and exercise raises myocardial O2 demand.

· Differentiate: 18 mm is well above the 12 mm or less of a trained heart; the septum is 18 mm against a posterior wall of 10 mm, so the hypertrophy is asymmetric rather than symmetric; the cavity at 42 mm is small rather than above 55 mm; relaxation is impaired rather than normal or enhanced; deep lateral T wave inversions and pathologic Q waves are not the isolated high voltage of an athlete; and the sudden death of a maternal uncle at 31 years fits a family history of the disease.

· Investigate: measure septal and posterior wall thickness, expecting 15 mm or more and asymmetric rather than 12 mm or less and symmetric; measure the cavity dimension, expecting below 45 mm rather than above 55 mm; assess diastolic filling, expecting impaired relaxation rather than normal or enhanced filling; read the electrocardiogram for deep T wave inversion and pathologic Q waves rather than isolated high voltage; look for dynamic outflow obstruction; take the family history for hypertrophic cardiomyopathy and sudden death; and note that the athlete's pattern would regress after about three months of detraining while this will not.

· Decide: dehydration from running in the heat lowered preload, and the heat also lowered afterload while catecholamines raised contractility and rate, each of which made the ventricle smaller and more forceful and worsened the outflow gradient. The risk of sudden death comes from reentrant ventricular arrhythmias arising in disarrayed, fibrotic myocardium.

· Justify: lower preload, lower afterload, greater contractility, and tachycardia all reduce ventricular size during systole, and a smaller cavity brings the septum and the mitral leaflet closer together, so obstruction worsens. Standing lowers preload and makes the murmur louder; a sustained handgrip raises afterload, enlarges the cavity, and makes it softer.

Distinctions that matter

· Chest tightness here is a supply and demand problem within an abnormal myocardium rather than epicardial stenosis.

· Maneuvers are read by their effect on ventricular size: smaller cavity means louder murmur in this condition and the opposite in regurgitant lesions.

Common misconceptions

· Attributing the findings to training because the patient is an athlete with a slow pulse. Asymmetry, the small cavity, impaired relaxation, the electrocardiogram, and the family history all point away from adaptation.

· Expecting the murmur to behave like a regurgitant murmur with handgrip. Raising afterload softens this one.

Where it is taught: Section C05, paragraphs on hypertrophic cardiomyopathy and dynamic outflow obstruction, the comparison table, and the stop and predict box of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C02-Q01

Board-style question: Accounting for the rise in oxygen uptake

Correct answer: B. An approximately fourfold rise in cardiac output together with an approximately threefold rise in O2 extraction

Why it is right: Both terms of the Fick equation rise: cardiac output goes from 5 to 21 L/min as heart rate and stroke volume increase, and extraction goes from 5 to 15 mL/dL as capillaries are recruited, tissue PO2 falls onto the steep part of the dissociation curve, and higher PCO2, H+, and temperature shift the curve to the right.

A is wrong: Arterial O2 content is 20 mL/dL at rest and 20 mL/dL at peak in this man, so pulmonary gas exchange contributed nothing to the rise.

C is wrong: The arteriovenous difference clearly changed, from 5 to 15 mL/dL, so attributing the rise to flow alone contradicts the numbers given.

D is wrong: Cardiac output rose from 5 to 21 L/min, so it did not stay at its resting value and extraction cannot be the only contributor.

A competent response contains

· VO2 equals cardiac output times the arteriovenous O2 difference.

· Cardiac output rose from 5 to 21 L/min, about fourfold, and the arteriovenous difference rose from 5 to 15 mL/dL, about threefold, so VO2 rose from about 250 mL/min to about 3.2 L/min.

Distinctions that matter

· Arterial content is held constant by healthy lungs, so the whole widening of the difference comes from the venous side.

Common misconceptions

· Expecting arterial O2 content to rise with exercise. It is already nearly fully saturated at rest.

Where it is taught: Section C02 of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C03-Q02

Board-style question: Regional flow during submaximal cycling

Correct answer: A. Coronary blood flow rises about three- to fourfold, because myocardial O2 extraction is already near maximal at rest

Why it is right: Resting myocardial extraction of about 70% to 75% leaves almost no extraction reserve, so the rise in myocardial O2 demand during exercise has to be met by a three- to fourfold rise in coronary flow, produced mainly by adenosine-mediated dilation.

B is wrong: Renal blood flow falls to about one quarter of its resting value at high workloads through sympathetic constriction and angiotensin II, and filtration fraction rises because flow falls more than GFR.

C is wrong: Cerebral blood flow is held nearly constant by autoregulation and PCO2 sensitivity, so its absolute flow stays about 0.75 L/min while its percentage share falls.

D is wrong: Splanchnic blood flow falls from about 1.4 to about 0.3 L/min, and splanchnic venoconstriction moves reservoir blood centrally, because sympathetic constriction dominates that bed.

A competent response contains

· The myocardium already extracts about 70% to 75% of delivered O2 at rest, so extra demand has to be met by raising flow.

· Coronary flow rises about three- to fourfold through mainly adenosine-mediated dilation, from about 0.25 to about 1 L/min.

Distinctions that matter

· A bed that already extracts nearly all the O2 it receives has no extraction reserve, so flow is its only route to meeting demand.

Common misconceptions

· Reading an unchanged percentage share of cardiac output as unchanged flow. The coronary share stays about 4% while flow quadruples.

Where it is taught: Section C03 of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C04-Q03

Board-style question: Hemodynamics of a sustained handgrip

Correct answer: D. Total peripheral resistance is unchanged or higher, and systolic and diastolic pressure both rise sharply

Why it is right: Rising intramuscular pressure restricts the contracting muscle's own flow, so metabolites accumulate and the metaboreflex with central command produces widespread sympathetic vasoconstriction; total peripheral resistance stays the same or rises while cardiac output rises modestly, so both systolic and diastolic pressure climb sharply and the ventricle faces a pressure load.

A is wrong: Falling resistance with a slightly lower diastolic pressure and a markedly higher stroke volume is the dynamic exercise pattern, not the response to a sustained handgrip.

B is wrong: A fourfold rise in cardiac output with a widened pulse pressure describes vigorous running; in static exercise output rises only modestly and mainly through heart rate.

C is wrong: Heart rate rises moderately during a handgrip rather than falling, because central command resets the baroreflex upward so the reflex does not oppose the rise in pressure.

A competent response contains

· A sustained isometric contraction compresses the muscle's own vessels, metabolites accumulate, and the metaboreflex with central command drives a strong sympathetic pressor response.

· Cardiac output rises modestly through heart rate, stroke volume changes little, and total peripheral resistance is unchanged or rises, so systolic and diastolic pressure both rise sharply.

Distinctions that matter

· The pressure response in static exercise comes from resistance failing to fall, while in dynamic exercise it comes from a large rise in output against falling resistance.

Common misconceptions

· Treating a handgrip as a small version of running. It loads the ventricle with pressure rather than volume.

Where it is taught: Section C04 of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C05-Q04

Board-style question: Reading an echocardiogram in a trained cyclist

Correct answer: C. The enlarged cavity with symmetric wall thickening and enhanced diastolic filling

Why it is right: An enlarged cavity above 55 mm with proportional symmetric thickening and enhanced early filling is the eccentric pattern of a repeated volume load, and it is the opposite of the small, stiff, asymmetrically thickened ventricle of hypertrophic cardiomyopathy.

A is wrong: A resting rate of 46/min reflects vagal tone and a large stroke volume, but bradycardia alone does not exclude hypertrophic cardiomyopathy.

B is wrong: Isolated high QRS voltage is common in trained athletes and can also occur with pathologic hypertrophy, so on its own it carries little weight.

D is wrong: A wall thickness of 11 mm sits below the 12 mm limit for an athlete, but thickness taken on its own leaves the gray zone unresolved and ignores the cavity and diastolic findings that decide the case.

A competent response contains

· Endurance training remodels the left ventricle eccentrically: an enlarged cavity, mild symmetric wall thickening, a normal wall-to-cavity ratio, an enlarged right ventricle and atria, and normal or enhanced diastolic filling.

· Hypertrophic cardiomyopathy gives a small or normal cavity, often below 45 mm, with asymmetric septal thickening of 15 mm or more and impaired relaxation.

Distinctions that matter

· Cavity size with diastolic function is the discriminator, because hypertrophic cardiomyopathy produces a small stiff cavity while training produces a large compliant one.

Common misconceptions

· Taking bradycardia or high voltage as evidence of adaptation. Both can appear in either condition and neither settles the question.

Where it is taught: Section C05 of the CV-12 Exercise and the Cardiovascular System notes.